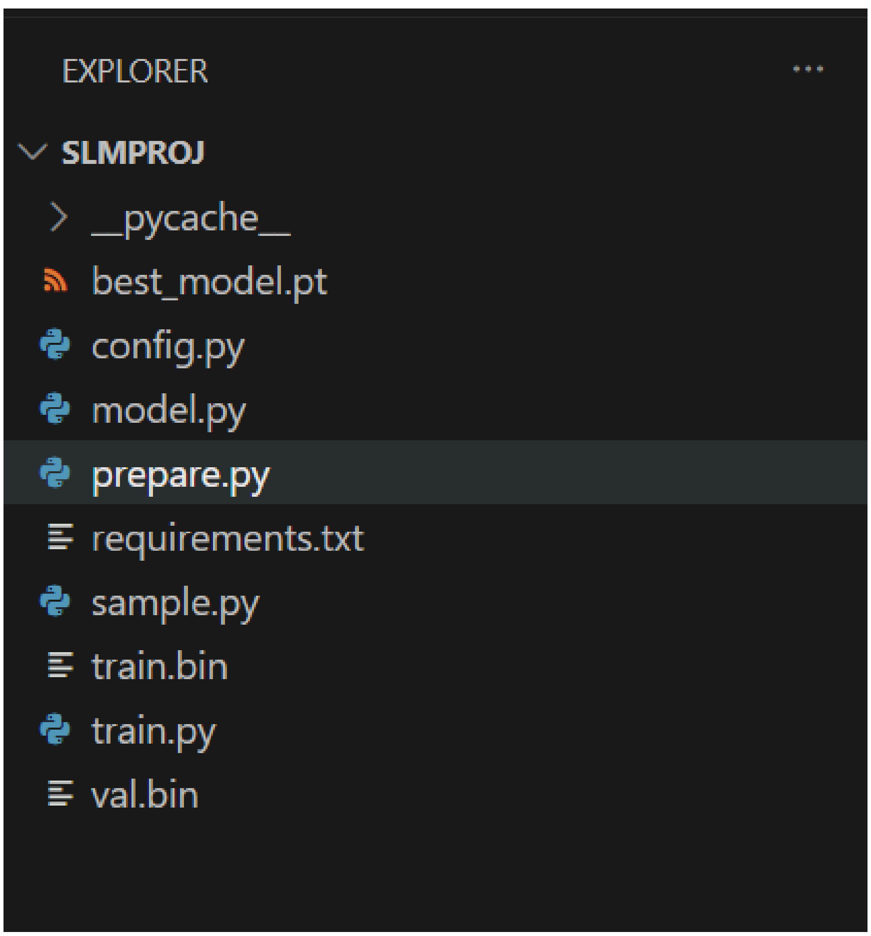
slm task 2 doc

Next-token prediction is where a model is trained to estimate the most likely subsequent unit of text (token) based on the preceding context.

A langage model takes tokenized text as input and predicts the probability of the next token. During training it learns language patterns by comparing its predictions to the actual next tokens and minimizing prediction error.

Tokenization is the foundational process in Artificial Intelligence that slices raw text into smaller, standardized units called tokens.Text is made into integers after normalising and converting it into a numerical token id that can be processed by nerual networks



The model consists of token embeddings, positional embeddings, transformer encoder layers, layer normalization and a linear language modeling head. The output shape is (batch_size, block_size, vocab_size).

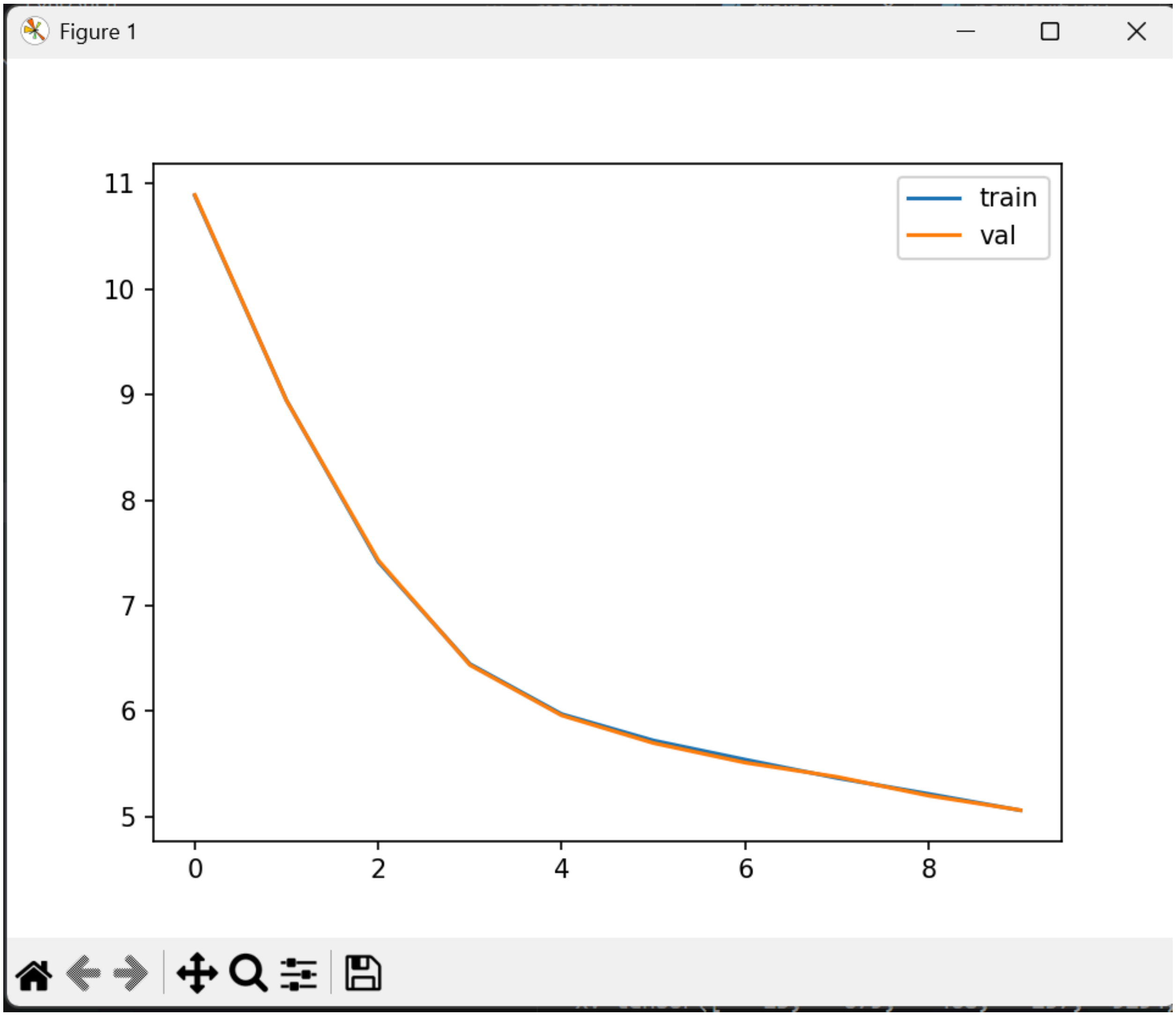
AdamW was used as the optimizer because it combines adaptive gradient updates with weight decay regularization.

and CosineAnnealingLR was used to gradually reduce the learning rate during training

The model was trained using cross-entropy loss. Training and validation losses were periodically evaluated and the best checkpoint was saved whenever validation loss improved.

```
PS C:\Users\avent\OneDrive\Desktop\slmproj> python train.py
iter 450 loss 5.6311
iter 450 loss 4.9853
iter 450 loss 5.2861
iter 450 loss 5.1628
iter 450 loss 5.2707
iter 450 loss 5.1781
iter 450 loss 5.2894
iter 450 loss 5.1845
iter 450 loss 5.4247
iter 450 loss 5.0955
iter 450 loss 5.0912
iter 450 loss 5.1567
iter 450 loss 5.1585
iter 450 loss 5.3206
iter 450 loss 5.2929
iter 450 loss 5.2431
iter 450 loss 5.3103
iter 450 loss 5.3098
iter 450 loss 5.1399
iter 450 loss 5.1224
iter 450 loss 5.0601
iter 450 loss 5.1426
iter 450 loss 4.9906
iter 450 loss 5.2486
iter 450 loss 5.0930
step 450 train 5.2080 val 5.1926
Training Complete
```

SHIFT TEST
x: tensor([13,   679,   468,   257, 9294,   286, 1660,   290,   734, 10912])
y: tensor([679,   468,   257, 9294,   286, 1660,   290,   734, 10912, 24314])



Perplexity was calculated as exp(validation_loss). Generated samples were evaluated qualitatively for grammatical correctness, coherence and ability to remain on topic.

```
d good
PS C:\Users\avent\OneDrive\Desktop\slmproj> & C:\Users\avent\Appl
ata\Local\Programs\Python\Python313\python.exe c:/Users/avent/On
Drive/Desktop/slmproj/perplexity.py
Validation Loss: 5.1926
Perplexity: 179.93577829922913
```

The trained model generated new stories from prompts such as "Once upon a time". Different temperature values were tested (1.0 and 0.7) to observe the tradeoff between creativity and predictability.

1.0 :

```
ata\Local\Programs\Python\Python313\python.exe c:/Users/avent/One
Drive/Desktop/slmproj/sample.py
Once upon a time " friend's and a gave mean the, The forever and
said,, and got, they to play ran the bear, and said, " their, "
eliber walked had you he to together. The girl fun, the boy coul
 asked and the waste, They named, so and the about and to to the
put in the looked a bright. She wanted


 his wanted to back First not said They saw a time, "ries.<|endo
text|>Once you she was so happy and he
```

0.7:

```
PS C:\Users\avent\OneDrive\Desktop\slmproj> python verify.py

Lily went to her mom and said, "Mom, I found this needle. Can you
 share it with me and sew my shirt?" Her mom smiled and said, "Ye
s, Lily, we can share the needle and fix your shirt."

Together, they shared the needle and sewed the button on Lily's s
hirt. It was not difficult for them because they were sharing and
 helping each other. After they finished, Lily thanked her mom fo
r sharing the needle and fixing her shirt. They both felt happy b
ecause they had shared and worked together.<|endoftext|>Once upon
 a time, there was a little car named Beep. Beep loved to go fast
 and play in the sun. Beep was a healthy car because he always ha
d good
```